



Python Programming

We will be starting shortly

Sit back and relax while you wait!

(Slides for today's lecture are on Scientia)



Quiz

Who is a snake's favourite author?

[answer given in lecture]



How was your week?





Done everything and need more exercises?

`https://edabit.com/challenges`

Otherwise, continue ploughing through the materials!



Your code works correctly!

Now what?



Writing efficient code

Improving the algorithm



Prime number validator (from Lesson 4)

100

Exhaustive search



Reduce search space?

Factors of 100:

2, 4, 5, 10, 20, 25, 50



Prime number validator (from Lesson 4)

100



Factors of 100:

2, 4, 5, 10, 20, 25, 50



Prime number validator (from Lesson 4)

100



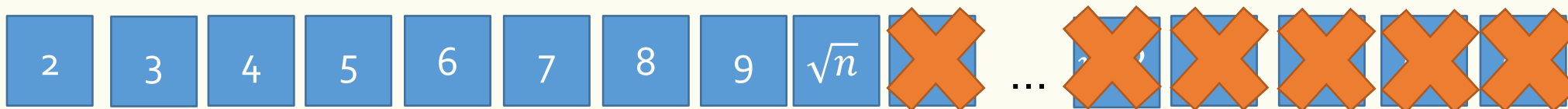
Factors of 100:

2, ~~4~~, 5, 10, 20, ~~25~~, 50



Prime number validator (from Lesson 4)

100



Factors of 100:

2, ~~4~~, 5, 10, 20, ~~25~~, 50



Prime number validator (from Lesson 4)

```
n = int(input("Please enter an integer: "))

divisor = 2
is_prime_number = True

while divisor < n:
    if n % divisor == 0:
        is_prime_number = False
        break
    else:
        divisor = divisor + 1

print(n>1 and is_prime_number)
```



Prime number validator (from Lesson 4)

```
n = int(input("Please enter an integer: "))

divisor = 2
is_prime_number = True

while divisor ≤ math.sqrt(n):
    if n % divisor == 0:
        is_prime_number = False
        break
    else:
        divisor = divisor + 1

print(n>1 and is_prime_number)
```



Square root estimator (from Lesson 4)

```
n = float(input("Please enter a positive number: "))

step_size = 0.000001
tolerance = 0.0001

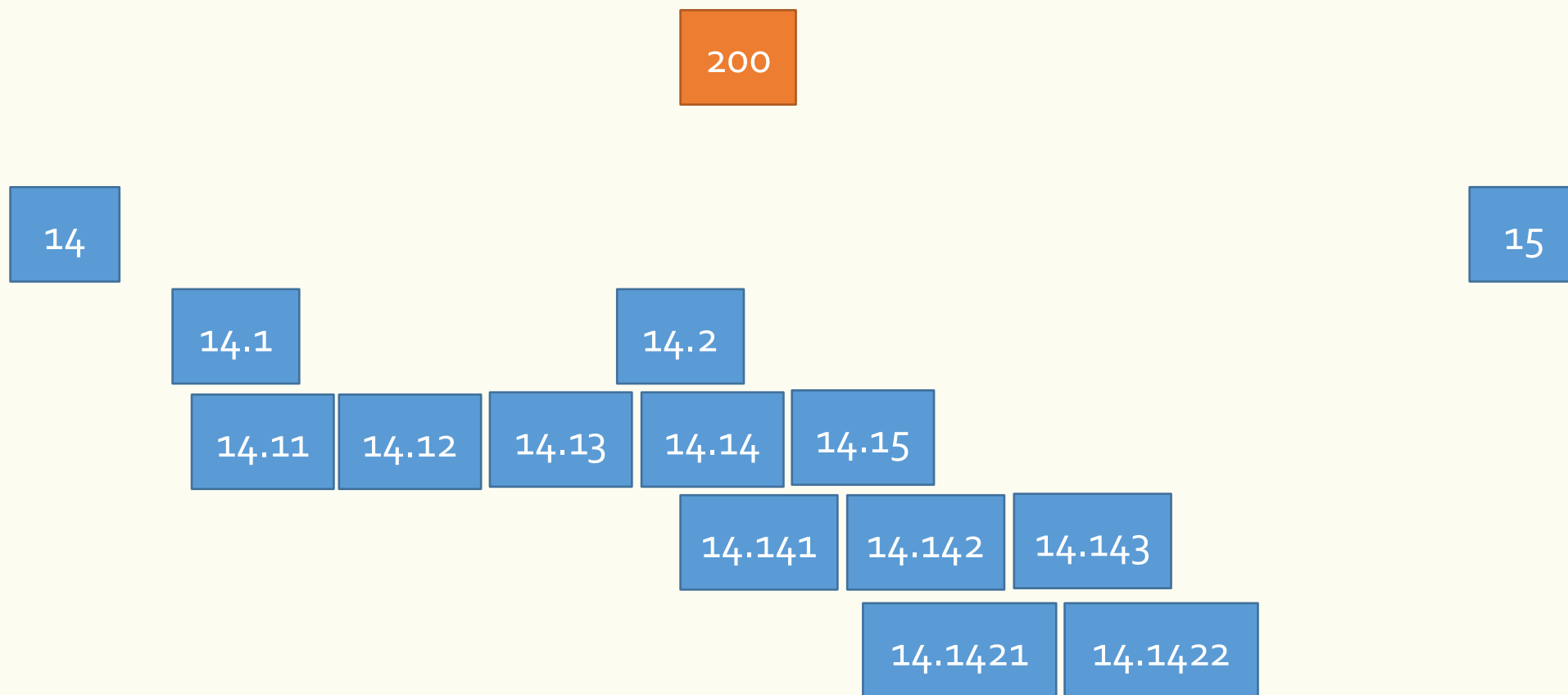
root = 0
while root ≤ n:
    if abs(root*root - n) < tolerance:
        print(root)
        break
    else:
        root = root + step_size
```



How would
you make this
more
efficient?



Square root estimator (from Lesson 4)





Writing efficient code

Using efficient operations



Performance tuning #1

- Concatenating a list of strings into a single string

```
items = ["a", "b", "c", "d", "e", "f", "g", "h",  
         "i", "j", "k", "l", "m", "n", "o", "p", "q", "r",  
         "s", "t", "u", "v", "w", "x", "y", "z"]
```

```
abcdefghijklmnopqrstuvwxyz
```



Performance tuning #1

- Which will be faster?

```
s = ""  
for item in items:  
    s += item
```

```
s = "".join(items)
```



Writing efficient code

Using optimal data structures



Performance tuning #2

- Filter a list of strings to return another list of unique strings

```
['269', '246', '324', '481', '324', '481', '687', '481', '612', '612']
```



```
['269', '246', '481', '324', '612', '687']
```



Performance tuning #2

```
filtered_items = []  
  
for item in items:  
    if item not in filtered_items:  
        filtered_items.append(item)
```

```
filtered_items = list(set(items))
```

```
filtered_items = list(dict(zip(items, [None]*len(items))))
```



Performance tuning #2

<filter_with_list>,	list size 10,	0.0008123 seconds
<filter_with_set>,	list size 10,	0.00037 seconds
<filter_with_dict>,	list size 10,	0.0007849 seconds
<filter_with_list>,	list size 100,	0.034291 seconds
<filter_with_set>,	list size 100,	0.0013613 seconds
<filter_with_dict>,	list size 100,	0.0031682 seconds
<filter_with_list>,	list size 600,	0.93456 seconds
<filter_with_set>,	list size 600,	0.011111 seconds
<filter_with_dict>,	list size 600,	0.021305 seconds
<filter_with_list>,	list size 4500,	11.232 seconds
<filter_with_set>,	list size 4500,	0.041099 seconds
<filter_with_dict>,	list size 4500,	0.12243 seconds
<filter_with_list>,	list size 30000,	91.623 seconds
<filter_with_set>,	list size 30000,	0.20946 seconds
<filter_with_dict>,	list size 30000,	0.79622 seconds



Performance tuning #3

- Is list comprehension faster? (Lesson 9)

```
s = []  
for item in items:  
    s.append(item)
```

```
S = [item for item in items]
```



Next week's schedule

Tue 2-3pm	Tue 3-4pm	Thu 4-6pm	Fri 11am-1pm
LAB 219	LECTURE 144	LAB 202/206	LAB 202/206

Lecture topic: Searching algorithms
CW₁ to be released on Tuesday



Any (anonymous) feedback for us?

- <https://forms.gle/NbaFwtvT2xhWbeSW7>

